2107540 Data Science for Environmental and Sustainability
Final Project Report
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Topic : Investment in Climate Adaptation Finance in Thailand
Article Link : https://rocketmedialab.co/climate-adaptation-finance/

Overview

The article *Climate Adaptation Finance in Thailand* examines how funding for climate adaptation is distributed across Thailand. The data were compiled by the Climate Finance Network Thailand (CFNT) in collaboration with the Puey Ungphakorn Institute for Economic Research (PIER). The article draws on the Thailand Climate Finance Tracker 2026, which covers both climate mitigation and adaptation investments from 2018 to May 2026.

The purpose of the article is to assess Thailand's financial preparedness for climate change by examining how much funding is available, where it is allocated, and which sectors receive the most investment. The article also aims to show whether Thailand's current investment patterns are sufficient for dealing with increasing climate-related risks, such as floods, droughts, and extreme weather events.

According to the article, Thailand received approximately 279.5 billion Thai Baht in climate adaptation finance during the period covered by the database. The data include investment from both public and private sectors, as well as different sources of funding within Thailand and from international organizations. This makes the article useful for understanding the overall financial landscape of climate adaptation in the country.

Data Analysis Techniques

This article uses descriptive data analysis to visualize climate adaptation finance in Thailand. The data are classified by location, investment sector, funding source, and number of projects. This allows the authors to compare the amount of funding between provinces and to identify which sectors receive the largest share of investment.

The report uses two main techniques to visualize insights. The first technique is a **Choropleth Map**, which is used to visualize how funding is distributed across Thailand. The funding amounts are abbreviated in million Thai Baht, allowing readers to compare investment levels between different provinces. This map makes it easier to identify areas that receive high levels of funding and areas that receive relatively low levels of investment.

The second technique is a **Treemap**, which is used to visualize the percentage of funding used by each government sector. The size of each section represents the amount of funding allocated to that sector. This makes it possible to see the difference between major sectors, such as water management, agriculture, and urban development, and smaller sectors, such as public health, infrastructure, and industry.

In addition to these visualizations, the article uses basic comparisons and rankings. For example, it compares the five areas with the highest levels of funding and the five areas with the lowest levels of funding. It also compares the number of projects with the total amount of funding. This is important because a province with many projects does not necessarily receive the highest total budget.

Results and Examples

The analysis found that since 2018 , Thailand has received approximately 279.5 Billion Thai Baht in climate adaptation finance of which the government sector is the main investor , around 4 out of 5 of climate adaptation fundings. Showing that the government has a strong important role on fundings. The top three provinces with highest budget are *Bangkok* with 1,239.99 Million THB , *Samut Prakan* with 992.33 Million THB and *Pattaya* with 304.53 Million THB respectively.

From the perspective of government sectors. The top three sectors with the highest budget are *Water Resources Management Sectors* with 168 Billion THB , as the most proportion amongst all sectors , followed by *Food and Agriculture Sectors* with 27.8 Billion THB (calculated as 11.8 percent) and *Urban and Habitations* with 21.4 Billion THB as 9 percent.

Overall, Thailand's climate adaptation budget is concentrated mainly on water management because the country regularly faces droughts and floods. However, recent flooding shows that funding and gray infrastructure alone may not guarantee effective adaptation and could even lead to maladaptation. Thailand should therefore combine gray infrastructure with green infrastructure and nature-based solutions. The government should also consider distributing adaptation funding more widely, as investments are currently concentrated in major economic areas that are highly vulnerable to the social and economic impacts of climate change.

Critical Review: Strengths and Limitations

The article's strengths are its use of recent data, clear classification of investment sectors, and effective visualizations such as the Choropleth Map and Treemap. The comparison between funding amounts and the number of projects also makes the analysis more informative.

However, the amount of funding does not necessarily show project effectiveness. Some data may be incomplete because not all investments are publicly disclosed. In addition, provinces have different population sizes and climate risks, so comparing total funding alone may not provide a complete picture. Further research should examine project outcomes and funding effectiveness.

(Summarized by Myself , Some written by ChatGPT and modified with AI)